

Input and Output Screenshots for Week 2 Citi Bike Example

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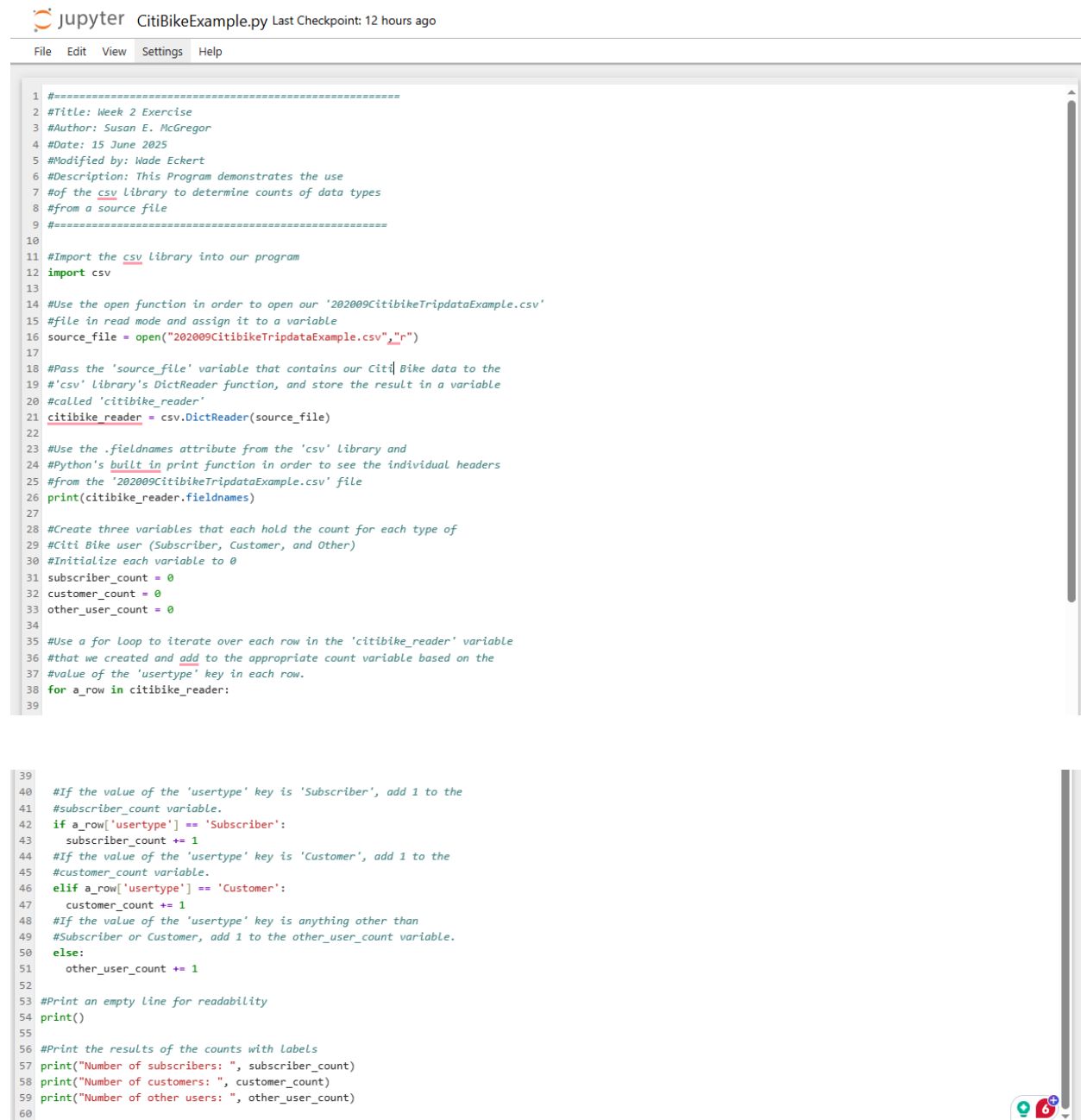
DSC 350: Data Wrangling for Data Science

Dr. David Kinney

June 15, 2025

Input and Output Screenshots for Week 2 Citi Bike Example

Input code, comments, and documentation:



```

1  #####
2  #Title: Week 2 Exercise
3  #Author: Susan E. McGregor
4  #Date: 15 June 2025
5  #Modified by: Wade Eckert
6  #Description: This Program demonstrates the use
7  #of the csv library to determine counts of data types
8  #from a source file
9  #####
10
11 #Import the csv library into our program
12 import csv
13
14 #Use the open function in order to open our '202009CitiBikeTripdataExample.csv'
15 #file in read mode and assign it to a variable
16 source_file = open("202009CitiBikeTripdataExample.csv", "r")
17
18 #Pass the 'source_file' variable that contains our Citi Bike data to the
19 #'csv' library's DictReader function, and store the result in a variable
20 #called 'citibike_reader'
21 citibike_reader = csv.DictReader(source_file)
22
23 #Use the .fieldnames attribute from the 'csv' library and
24 #Python's built in print function in order to see the individual headers
25 #from the '202009CitiBikeTripdataExample.csv' file
26 print(citibike_reader.fieldnames)
27
28 #Create three variables that each hold the count for each type of
29 #Citi Bike user (Subscriber, Customer, and Other)
30 #Initialize each variable to 0
31 subscriber_count = 0
32 customer_count = 0
33 other_user_count = 0
34
35 #Use a for Loop to iterate over each row in the 'citibike_reader' variable
36 #that we created and add to the appropriate count variable based on the
37 #value of the 'usertype' key in each row.
38 for a_row in citibike_reader:
39
40     #If the value of the 'usertype' key is 'Subscriber', add 1 to the
41     #subscriber_count variable.
42     if a_row['usertype'] == 'Subscriber':
43         subscriber_count += 1
44     #If the value of the 'usertype' key is 'Customer', add 1 to the
45     #customer_count variable.
46     elif a_row['usertype'] == 'Customer':
47         customer_count += 1
48     #If the value of the 'usertype' key is anything other than
49     #Subscriber or Customer, add 1 to the other_user_count variable.
50     else:
51         other_user_count += 1
52
53 #Print an empty line for readability
54 print()
55
56 #Print the results of the counts with labels
57 print("Number of subscribers: ", subscriber_count)
58 print("Number of customers: ", customer_count)
59 print("Number of other users: ", other_user_count)
60

```

Output of code:

```
python3 mai... Ask Assistant 409ms • Just now ✓  
['tripduration', 'starttime', 'stoptime', 'start station id', 'start station name', 'start station latitude', 'start station longitude', 'end station id', 'end station name', 'end station latitude', 'end station longitude', 'bikeid', 'usertype', 'birth year', 'gender']  
Number of subscribers: 58961  
Number of customers: 17713  
Number of other users: 0
```